Sacha Loi

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EDUCATION

University of Warwick — **Warwick, UK**
Bachelor of Science (with Honours): Computer Science — Sep 2025 – Jul 2028 (expected)
- Average Grade: 71.1% (First Class)

Monk's Walk, Welwyn Garden City — **Hertfordshire, UK**
Secondary Education — Sep 2018 – Jul 2025
- A-Levels: Computer Science (A*), Physics (A*), Mathematics (A)
- 10 GCSEs: Grades 7–9 including Computer Science and Mathematics

WORK EXPERIENCE

Splunk — **London, UK**
Software Engineering Work Experience — Aug 2024
- Collaborated within a Scrum-based software engineering team on a live commercial project, contributing to sprint planning, stand-ups, and code review discussions throughout the SDLC
- Deployed and configured a standalone Splunk Enterprise instance, ingesting and indexing over 159,000 log events across multiple source types to explore real-world applications in IT operations and security monitoring
- Employed SPL (Splunk's search language) to query, aggregate, and visualise the data, building a multi-panel dashboard covering web traffic and authentication activity

ServiceNow — **London, UK**
Technical Consulting Work Experience — Jul 2024
- Built working knowledge of core ServiceNow platform architecture, including ITSM, CMDB, and NowAssist GenAI
- Developed and tested Client Scripts (JavaScript), Business Rules (JavaScript), and UI Policies in sub-production developer instances
- Supported enterprise ServiceNow deployments through platform configuration and requirements gathering, in addition to participating in Agile sprint ceremonies under the NowCreate methodology

Morningstar — **London, UK**
Software Engineering Work Experience — Jun 2021
- Engineered and debugged a Python script that queried a company database to automate dataset generation, streamlining statistical analysis for other departments
- Participated in daily stand-ups and sprint planning, gaining hands-on exposure to Agile implementation methodologies through the use of JIRA to plan, track, and deliver work efficiently

ADDITIONAL EXPERIENCE

Brocket Hall — **Hertfordshire, UK**
Front of House — May 2026 – Current
- Led and worked events with over 100 people such as weddings, corporate events and funeral wakes in a diverse range of roles including bar, food running and taking orders to a high-end service standard
- Ran the clubhouse in short stints to support the kitchen, bar and food running for a smooth operation and delivery

PROJECTS

3D Snake on an 8×8×8 LED Cube | *C, STM32, libopencm3* (team of 2)
- Co-developed a 3D Snake game in C for an STM32 microcontroller, driving a 512-LED cube over USART with 3D movement, wall and self-collision detection, and random apple placement that avoids the snake's body
- Compressed each frame into a single 65-byte USART packet by storing every LED row as a bitmask and sorting rows into transmission order

Fitness Tracker Web App | *Python, Flask, SQLAlchemy, SQLite, Jinja*
- Built a full-stack Flask app that calculates BMI, BMR (Mifflin–St Jeor) and TDEE and turns the user's goal into daily calorie and macronutrient targets
- Designed a relational schema for users, foods and daily meal logs, with Alembic migrations and a 330+ item nutrition dataset; each user's remaining macronutrient targets automatically reset each day

Maze-Solving Robot Controller | *Java*
- Engineered an autonomous controller that navigates unseen mazes, classifying corridors, junctions and dead ends and backtracking with a stack
- Recorded the optimal exit at each junction so subsequent runs follow the most efficient route to the target from any start position

Proof-Language Parser | *Python*
- Implemented a deterministic pushdown automaton that parses proofs, using the stack to match nested sub-proofs
- Validated 8 inference rules (AND, OR, IMPLIES, NOT) with scope tracking for formulas derived under assumptions

SKILLS & INTERESTS

Technical Skills:	Python, Java, HTML/CSS, JavaScript, C#, SQL, C
Frameworks & Tools:	Flask, SQLAlchemy, Unity, Gradle, Git/GitHub, Splunk (SPL), ServiceNow Platform, JIRA
Leadership/Volunteering:	St Albans Youth Talk (Fundraising), Leading local litter-picking team, Mentoring and Teaching BBC Micro bits, Duke of Edinburgh Bronze
Sports & Other:	Cricket, Jeet Kune Do (JKD London), Golf (Handicap 26), 5-a-side active leagues (Football), Warwick Squash Society, Warwick Computing Society, Full UK Driving Licence